

SREERAG E G

RPA / Automation Engineer

Python | Process Automation & Orchestration | AWS (ECS/SQS)

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PROFESSIONAL SUMMARY

RPA / Automation Engineer with 1+ year of production experience designing distributed, HIPAA-compliant process-automation infrastructure for US Healthcare RCM workflows. Expert in Python, FastAPI, and AWS (ECS/SQS) for bot orchestration at scale, with hands-on experience building 130+ production automations (bots) covering claim submission, EOB processing, claim retrieval, patient enrollment, and FAP data collection across 40+ commercial insurance portals.

CORE TECHNICAL STACK

- **RPA & Process Automation:** Playwright, Selenium, Web Scraping, PDF/Structured Data Extraction, Bot Orchestration.
- **Resilience Engineering:** Page Object Model (POM), Error Classification & Self-Healing Logic, Retry & Fallback Mechanisms.
- **Backend & Orchestration:** FastAPI, Django, REST API Design, Celery, Message-Queue-Driven Async Processing.
- **Cloud & Infrastructure:** AWS (ECS, SQS, EC2, S3, Lambda), Docker, Git, CI/CD Pipelines.
- **Languages:** Python (Advanced OOP & Framework Design), SQL, JavaScript, Java.
- **Domain Specialization:** US Healthcare Insurance, Revenue Cycle Management (RCM), End-to-End Claims Lifecycle (Submission, Status, EOB/835, Patient Enrollment), HIPAA Compliance, CMS-1500.

PROFESSIONAL EXPERIENCE

Automation Engineer (RPA / Python) | Kalki Safalya Consulting Services LLP

May 2025 – Present

- **Bot Orchestration Gateway:** Developed a high-performance **FastAPI** orchestration layer that dynamically routes **100+ daily** bot executions to the correct automation based on incoming JSON payloads.
- **Scalable Bot Infrastructure:** Deployed a containerized worker pool on **AWS ECS**, using **AWS SQS** as a message broker to process **500+** claim-submission bot runs asynchronously per week.
- **Process Automation at Scale:** Built and shipped **130+** production bots (enrollment, claim submission, claim retrieval, consent, FAP data collection) across **40+** commercial insurance portals, increasing claim submission throughput by **40%** and eliminating **80+** hours of manual data entry monthly.
- **Self-Healing & Compliance:** Engineered automated error classification and self-healing bot logic to cut maintenance overhead by **30%**, with sensitive-data masking to maintain strict HIPAA compliance.
- **Reconciliation Reporting:** Built data tracking and reconciliation dashboards in **Pandas** to cut billing discrepancies by **25%**, integrating RPA bots with enterprise Rails backends via message queues.

Full Stack Developer (Intern) | Aspire Systems

Nov 2024 – May 2025

- **API Engineering:** Developed and optimized **12+** REST API endpoints using **Django Rest Framework (DRF)** and MySQL, improving database query response times by **20%** for enterprise applications.
- **Process Reliability:** Automated API documentation using Swagger and supported code reviews for 15+ pull requests, reducing cross-team integration time.
- **Auth Systems:** Implemented secure OAuth 2.0 authentication flows (including Google OAuth) and CRUD operations, onboarding **200+** internal users to the enterprise portal.

Freelance RPA / Python Developer | Self-Employed

May 2024 – Nov 2024

- **Legacy Bot Migration:** Restructured and converted **30+** legacy Selenium bots into a modular, reusable Page Object Model (POM) framework, reducing bot failure rates by **50%**.
- **Third-Party Integration:** Integrated external REST APIs into bot workflows for automated data verification, accelerating form-processing speed by **15%**.

KEY PROJECTS

- **Structured Claim-Data Extraction & Reporting Pipeline (CPF & Claim Retrieval):** Built schema-driven JSON extraction bots (replacing OCR-on-PDF) across **11+** claim-payment-fetch and **7+** claim-retrieval portal integrations, using a canonical **ClaimRecord** envelope, HTML-table pagination scraping, and automated payment/status normalization.

EDUCATION & CERTIFICATIONS

- **B.Tech in Computer Science and Engineering** | Christ College of Engineering (7.85 CGPA).
- **Certifications:** Django for Web Development, Data Engineering with PySpark, IBM Python for Data Science, AI For Everyone (Andrew Ng).